

Flying with SeaLabMOS Systems (Airport Security Considerations) v2026.06.12

Budget an additional hour to go through security

It is likely that you will end up talking with a member of the TSA's Bomb Squad. In my experiences at different airports (SFO, HNL, ORD, EWR), they are the most relaxed person to explain things to. Be prepared to talk with the TSA agents at the gate, the Terminal Manager, the police, and maybe even the Airline Representative, all of whom have tended to be less friendly than the Bomb Squad. In my experience, it depends on the airport, the date, and the time.

Arriving and putting your bag through security

1. Do not remove the system from your bag when at the airport and when going through security. It might scare the TSA if you remove it from your bag before the scanner (ask me how I know). Place your bag in the bin / on the conveyor belt and tell the TSA agent standing there that you
 - a. Have a scientific device consisting of electronics inside a PVC tube that will look weird on the scanner, and
 - b. That they should call over their supervisor to take a look at it
2. Go through and collect the rest of your belongings, then wait for them to talk to you

At this point, they will start asking you what it is. Nothing bad should happen; calmly explain who you are and what the system is.

Explaining the system

It is an open source oceanographic sampling system (CTD, pressure / temp logger, etc.), considering entirely of consumer-off-the-shelf parts (electronics that anyone can easily purchase online), used to measure things like temperature in the ocean. I recommend trying to split the difference between technical and general language.

When the Bomb Squad person arrives and wants to know about the system itself, explain that it uses

1. Adafruit RP2040 Adalogger MCU and Lithium Ion 3.7V Battery, also from Adafruit
2. Real time clock FeatherWing to record time
3. Sensors - temperature, pressure, salinity, etc. as your system is built

You can pull up my website for images of other systems (also has my contact info):

<https://www.sophiescopazzi.com/projects/sealabmos>

Considerations

1. Don't make any sudden moves for the bag containing the system or even your own baggage that they screened, and you collected. Follow all of their directions. Stay relaxed and talk to them about what the system is and what it is for
2. You didn't put the system in your checked bag because you wanted to be on hand to explain what it is – for this exact reason *gesture at everyone gathering*
3. The TSA and/or the police may ask to take a photo of your identification and ticket. They've done this every time I've flown with the systems and nobody has said anything after

After the Bomb Squad person clears your device for flight, the next issue the Terminal Manager or Airline Representative might bring up is what they have termed "optics". Even though it's safe to fly, it can still look scary. As explained to me, "If you stood up on the plane with it in your hands and said something, people would take you seriously." If optics is brought up, explain that you fully understand and will not be removing it from your baggage until you reach your destination. If this doesn't satisfy them, offer that they can wrap it in a trash bag a TSA tape (I've had to do this once).